High Level Data Representations For HDM-4

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Background

Economic decision making is vital in highway management. The decisions deal with resource allocation to the most beneficial projects. Computer programs are vital tools in this decision making process.

The Highway Development and Management Model (HDM-4) is a popular appraisal tool for road projects. However adoption of HDM-4 is hindered by the large data requirements to set up and run analyses. High level data representations are a possible solution to these challenges. This project assessed high level data representations for the HDM-4 Model as implemented in HDM-Sentry, a new software aimed at increasing accessibility of the HDM-4 Models

HDM-Sentry (Simplified Entry)

- HDM-Sentry is a '*simplified HDM-4 data-preparation and analysis tool*' developed as a simpler and improved user interface to the HDM-4 model.
- Enables the user to set up and run an analysis in four simple steps
- Relies on abstraction of the data required by the HDM-4 workspace into simpler high level data descriptors

Objectives

- To assess the representative data descriptors for a typical section run in HDM-Sentry;
- Conduct a sensitivity analysis of the economic and deterioration model outputs to the data inputs for HDM-Sentry;
- Identify the significant high level inputs to HDM-Sentry;
- Investigate variations in the output across the input space;
- To investigate the applicability of the data descriptors in real world situations.

Methodology

Identifying the HDM-Sentry Inputs

- Four main inputs to HDM-Sentry were considered
  - **Traffic Volume**
  - **Traffic Growth**
  - **Initial road Condition**
  - **Geometry**

Full Factorial Experiment Design

- All levels of each variable combined with all levels of all the other variables and results recorded

Sensitivity Analysis

- Sensitivity analysis of NPV and average IRI in relation to the high level inputs

Comparison to real network

- High level inputs compared to the actual inputs from a real road network from Uganda National roads Authority (UNRA)

Input Variables

MENU	VARIABLES	CODE
Traffic Growth	None	1
	Low	2
	Medium	3
	High	4
Condition	New	0
	Good	1
	Fair	2
	Bad	3
	Poor	4
Traffic Volume	Low Traffic	0
	Medium Traffic	1
	High Traffic	2
Road Geometry	Straight and Level	0
	Mostly straight and gently undulating	1
	Bendy and generally level	2
	Bendy and gently undulating	3
	Bendy and severely undulating	4
	Winding and gently undulating	5
	Winding and severely undulating	6

Four input variables chosen with their respective levels turned into codes for input into the automated HDM-Sentry script.

Experiment Input Matrix

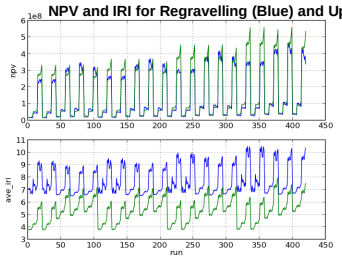
Run No.	INPUT PARAMETERS			
	Tg	Co	Tv	Ge
T2Run1	1	0	0	0
T2Run2	1	0	0	1
T2Run3	1	0	0	2
.	.	.	.	.
.	.	.	.	.
T2Run150	2	2	0	2
T2Run151	2	2	0	3
T2Run152	2	2	0	4
T2Run250	3	1	2	4
.	.	.	.	.
.	.	.	.	.
T2Run350	4	1	1	6
T2Run351	4	1	2	0
T2Run419	4	4	2	5
T2Run420	4	4	2	6

Each trial run of the full factorial experiment had 420 runs with different each run having different levels of input parameters.

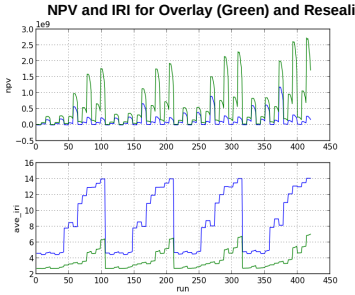
Results of ANOVA Sensitivity Analysis

- Sensitivity analysis show that for project appraisal of a road section using HDM-Sentry, the Traffic volume is the dominant input factor in relation to NPV.
- The road condition was also found to be a significant contributor to the NPV and average IRI for projects on bituminous surfaced roads. The effect of the condition on NPV was observed to depend on the maintenance alternative considered.
- The interaction of the condition and traffic volume inputs was observed to be significant to the NPV for bituminous surfaced roads.

Results



- Results for gravel surface showed a high response to traffic volume
- Significant differences between the peak NPV values and the rest of the values



- Results for bituminous surface showed a high response to both traffic volume and initial condition;
- Significant differences between the peak NPV values and the other values

Findings

- Traffic volume is the main input affecting the NPV and IRI for gravel roads;
- Traffic volume and Initial Condition are the main inputs affecting NPV and IRI for bituminous surfaced roads;
- There are significant interactions between traffic volume and condition;
- The output function for NPV shows large changes between the medium traffic and high traffic levels;
- The ranges for traffic volume and condition do not cover the whole possible input space when compared with the UNRA network.

Conclusions and Recommendations

- Sensitivity analysis shows that for project appraisal of a road section using HDM-Sentry, the Traffic volume is the dominant input factor in relation to NPV.
- The road condition was also found to be a significant contributor to the NPV and average IRI for projects on bituminous surfaced roads. The effect of the condition on NPV was observed to depend on the maintenance alternative considered.
- The interaction of the condition and traffic volume inputs was observed to be significant to the NPV for bituminous surfaced roads.
- Further work required on ways of reducing the magnitude of the discontinuities in the NPV output
- More work needed to ensure the input ranges of the high level data representations cover the possible values expected in practice.